

119. A 7-DAY MEDIUM INTENSITY BLOCK MICROCYCLE BOOSTS AEROBIC ADAPTATIONS AND PERFORMANCE!

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An important note before reading this KIW Research Note:

- Low intensity training: below the first threshold;
- Medium intensity training: between the two thresholds;
- High intensity training: above the second threshold;

When looking at the annual training plan, **classic linear periodization**—a progressive increase in volume first, followed by intensity (while maintaining or slightly reducing volume)—is the most commonly used and evidence-based strategy among successful endurance athletes. However, for **time-crunched cyclists**, this approach is not always feasible. Many can train only about **1 hour from Monday to Friday**, with the possibility of doing longer rides only on some weekends. As a result, they lack the time needed to progressively increase volume and properly build a classic linear periodization. Consequently, many of these cyclists introduce

a substantial amount of **medium- and high-intensity training early in the season**. This leads them to quickly reach a plateau in both volume and intensity—which if then maintained for several months **without any progressive overload**, increases the risk of stagnation.

In previous KIW Research Notes, we saw that for these athletes an effective alternative could be **high-intensity (HIT) block periodization/microcycles**—that is, concentrating HIT sessions over a few consecutive days (5-7) followed by recovery days or weeks, rather than distributing 2–3 HIT workouts evenly every week throughout the training plan. Specifically, we saw that:

1. When training volume is kept constant at around 8/10 hrs per week. **12 weeks of HIT block periodization lead to superior performance and physiological adaptations compared to an equally distributed HIT approach.** ([Learn more here](#))
2. A **6-day HIT shock microcycle** followed by **6 days of active recovery** is effective in improving cycling performance and so may serve as an **optimal tapering strategy**.

But does the same hold true for medium intensity?

Can a medium intensity block-microcycle increase improve performance more than regular training?

A study published by Mølmen and Colleagues (University of Inland Norway, Lillehammer) in August 2025 on *Medicine & Science in Sports & Exercise* tried to answer this question. (1)

WHAT DID THEY DO?

- 30 male professional or highly trained cyclists (age ~20, VO2max ~71) were recruited.
- **All participants completed both a 15-day regular training block (REG) and a medium-intensity training block (MIT), in a randomized order.** This means that participants were randomly assigned to start with either the regular training block or the medium-intensity block, with roughly half beginning with one condition and half with the other. Between the test at the end of each block and the start of the next one, participants rested or performed light training for two to three days.

MIT BLOCK

It consisted of six interval sessions at medium intensity (MIT) distributed over 7 days followed by a 6-day active recovery period:

DAY	Workout
DAY 1	7 × 10min MIT with 2.5min recovery
DAY 2	6 × 12min MIT with 3min recovery
DAY 3	5 × 14min MIT with 3.5min recovery
DAY 4	Rest Day
DAY 5	7 × 10min MIT with 2.5min recovery
DAY 6	5 × 14min MIT with 3min recovery
DAY 7	6 × 12min MIT with 3.5min recovery
DAY 8	complete rest
DAY 9	complete rest or 20–40 min of low-intensity
DAY 10	0.5–1.5 h of low-intensity exercise
DAY 11	20min LIT + 2 × 5-min MIT + 1min HIT with progressive intensity + 10min LIT
DAY 12	complete rest
DAY 13	20min LIT + 2 × 5-min MIT + 3X1min HIT with progressive intensity + 10min LIT

- What was the MIT intensity? MIT intervals were prescribed using an **RPE (Rating of Perceived Exertion) of 14–15** (“somewhat hard” to “hard”) on the Borg 6–20 scale. This corresponded to about **85% of maximum heart rate** and a blood lactate concentration of **roughly 2.8 mmol/L**. Therefore, it is very likely that the actual exercise intensity fell between the first and second physiological threshold.
- *“All MIT sessions were initiated by a standardized 15-min progressive warm-up protocol succeeded by 5 min of rest before starting the first work interval. Each session was finalized with 5 min of active recovery” (1), so they all lasted around 1h30min.*

REGULAR TRAINING BLOCK:

During the Regular Training Block, participants continued following their usual training plan, which included:

- 2x week MIT sessions, or 1x MIT session + 1xHIT session per week.

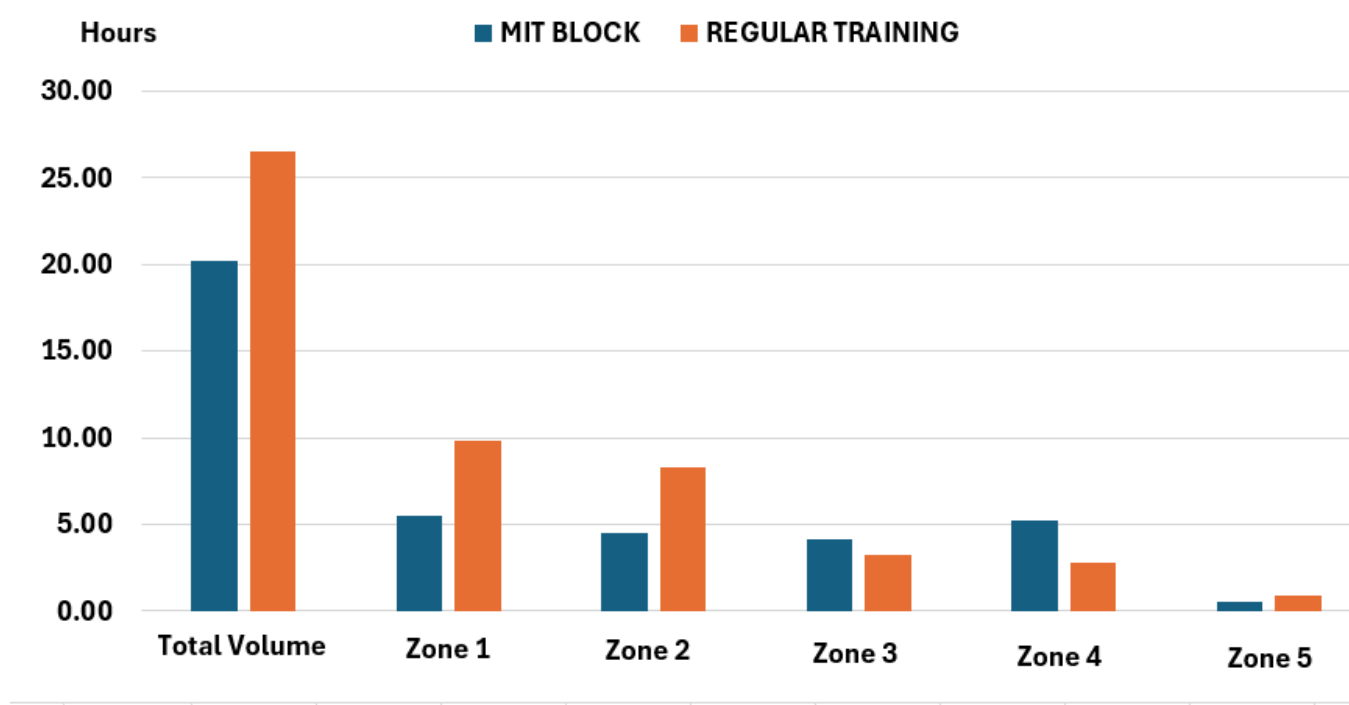
MIT sessions involved interval workouts consisting of 4–8 × 6–20 minutes performed at a Borg RPE of 14–16.

HIT sessions generally consisted of 4–6 × 4–8 minutes, performed at a Borg RPE of 16–19.

- Long endurance rides lasting 3–4 hours.
- A higher overall training volume compared to the MIT block.
- Additionally, “those who routinely engaged in resistance training were permitted to continue their sessions as usual” (1).

OTHER POINTS ABOUT THE STUDY DESIGN:

- The final two days of the protocol before the physiological and laboratory tests were kept easy and identical for both the MIT and the Regular Training block, in order to minimize any fatigue-related influence on the study results.
- Here below the differences in total training volume and time spent in each intensity zone between the MIT block and the Regular Training block:



Data from Mølmen et al. 1. The intensity zones in the graph are based on % of the mean maximum heart rate on 40min (HR40min) Zone 1 (<55% of HR40min. Zone 2 (56%–75% of HR40min. Zone 3 (76%–90% of HR40min. Zone 4 (91%–105% of HR40min. Zone 5 (>106% of HR40min.

WHAT DID THEY FIND?

The MIT block produced greater aerobic physiological and performance improvements compared to the Regular Training (REG) block:

- **VO₂max:**

MIT Block: from 70.1 to 71.4 ml/min/kg: +1.9%

REG: from 70.8 to 70.8 ml/min/kg: 0%

- **Power at 4 mmol/L of blood lactate concentration:**

MIT BLOCK: from 4.05 to 4.21 W/Kg: +4%

REG: from 4.16 to 4.11 W/Kg: -1.2%

- **Cycling Efficiency**

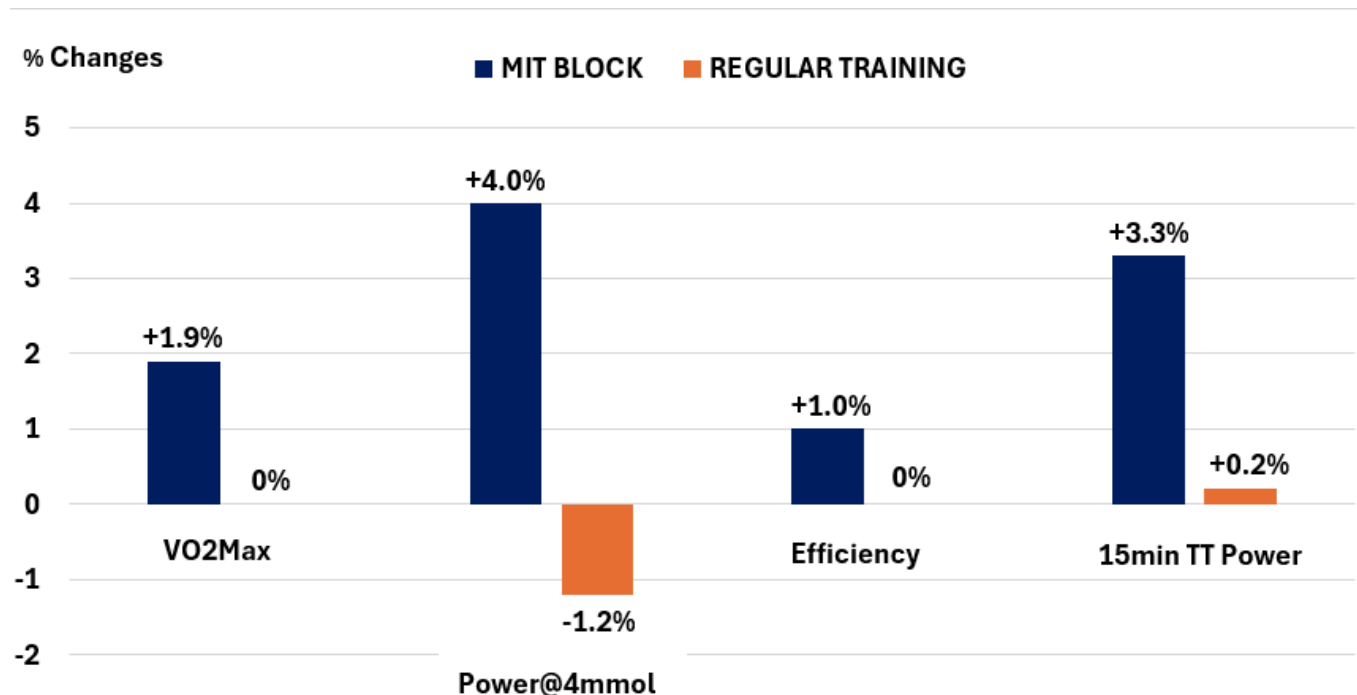
MIT BLOCK: from 19.4 to 19.6%: +1%

RE: from 19.4 to 19.4: 0%

- **Mean Power Output 15 min Time Trial**

MIT BLOCK: from 4.22 to 4.36 W/Kg: +3.3%

RE: from 4.30 to 4.31 W/Kg: +0.2%



Data from Mølmen et al. 2025 (1).

PRACTICAL TAKEAWAYS:

A Medium Intensity one week block microcycle followed by 6 days active recovery induces greater aerobic physiological and performance adaptations when compared to a time matched period of regular training.

GABRIELE'S COMMENT

I found these data are particularly interesting because they suggest that, in addition to high-intensity block microcycles, the same block periodization principle applied at medium intensity can also enhance aerobic adaptations and performance. **In other words, condensing a training stimulus into a few days—a “block/shock” microcycle—can be effective when using both HIT and MIT.**

Here below some considerations to keep in mind when putting into practice MIT block periodization:

- **This approach can be particularly useful for time-crunched cyclists**, who often cannot follow a classic linear periodization without maintaining similar volume and intensity for several weeks, which can lead to performance stagnation.
- **However MIT block periodization could also be applied to professional or non-time-crunched cyclists.** After building a base with progressive increases in volume during winter, MIT or HIT block periodization can be used. In this scenario, total training volume during the MIT/HIT block should be reduced compared to usual, but it is probably best to keep it higher than what was used in this study.
- Since a MIT block microcycle is likely to put a high level of strain on the body, it is important to **start the block fresh and well-rested**, so that the body can fully absorb the load.
- A strategy which can be useful to manage the high volume of medium-intensity (or high-intensity) training during block periodization is to include double sessions instead of a single daily session. As shown in a previous KIW research note, this can help to maintain intensity while reducing perceived fatigue and muscle soreness. However, if developing durability is also a goal, double sessions should be limited to once per week, in order to also practice sustaining effort under fatigue.

In the next KIW Research Notes, we will see more about medium vs high-intensity block microcycles in terms of aerobic adaptations and performance!

Thanks for Reading! Share with your Team!

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REFERENCE:

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1. Mølmen KS, Odden I, Sørensen MD, Rustaden AM, HAMMARSTRÖM D, Hansen J, Nygaard H, Hamarsland H, Rønnestad BR. A Moderate-Intensity Interval Training Block Improves Endurance Performance in Well-Trained Cyclists. Med Sci Sports Exerc. 2025 Aug 1;57(8):1780-1789.



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